

12.20 The vector potential in the superconductor, by continuity and regularity, must have the form

$$A_y = a' e^{-x/\lambda_L - i\omega t},$$

where a' can be determined by boundary condition, but we don't care here. Then,

$$\vec{E} = -\frac{1}{c} \frac{\partial \vec{A}}{\partial t} = \hat{y} i\omega A_y / c, \text{ and } \vec{B} = \nabla \times \vec{A} = \hat{z} \frac{\partial A_y}{\partial x} = -\hat{z} \frac{A_y}{\lambda_L}.$$

$$\text{Thus } Z_S = \frac{4\pi}{c} \frac{E}{B} = -i \frac{4\pi}{c} \frac{\omega}{\lambda_L} = -i \frac{4\pi}{c} \frac{2\pi}{\lambda} \lambda_L = -i \frac{8\pi^2}{c} \frac{\lambda_L}{\lambda}.$$

12.21. (a) From Eq. (7.58), with the substitution, the conductivity is

$$\sigma = \sigma_N + \sigma_S = \frac{n_N e_N^2}{m_N (\gamma_N - i\omega)} + \frac{n_S e_S^2}{m_S (\gamma_S - i\omega)} = \frac{n_N e_N^2}{m_N (\gamma_N - i\omega)} + i \frac{n_S e_S^2}{m_S \omega},$$

With $\gamma_S = 0$. In the limit $\omega \rightarrow 0$, the conductivity is dominated by the imaginary part.

(b) From Ohm's law, $\vec{J} = \sigma \vec{E}$. Assuming harmonic dependence, the Maxwell equation becomes

$$\nabla \times \vec{E} = -\frac{1}{c} \frac{\partial \vec{B}}{\partial t} = \frac{i\omega}{c} \vec{B}, \text{ and } \nabla \times \vec{H} = \frac{4\pi}{c} \vec{J} + \frac{1}{c} \frac{\partial \vec{D}}{\partial t} = -\frac{i\omega}{c} \left(1 + i \frac{4\pi\sigma}{\omega}\right) \vec{E} \text{ Thus,}$$

$$\nabla \times (\nabla \times \vec{E}) = \frac{i\omega}{c} (\nabla \times \vec{B}) = \frac{i\omega}{c} (\nabla \times \vec{H}) = \frac{i\omega}{c} \left(-\frac{i\omega}{c} \left(1 + i \frac{4\pi\sigma}{\omega}\right)\right) \vec{E} \text{ Since } \sigma \approx i \frac{n_S e_S^2}{m_S \omega} \text{ in the}$$

static limit by part (a), we have

$$\nabla \times (\nabla \times \vec{E}) = \nabla(\nabla \cdot \vec{E}) - \nabla^2 \vec{E} = -\frac{4\pi n_S e_S^2}{m_S c^2} \vec{E},$$

where we can identify the penetration as

$$\lambda_L = \left(\frac{m_S c^2}{4\pi n_S e_S^2} \right)^{1/2},$$

agreeing with Eq. (12.100).